

Summary

1. Data Collection

- Imported the Student Performance dataset into Python for analysis.
 - Loaded the dataset using Pandas to understand the available features and student records.
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2. Data Understanding

- Explored the dataset using functions like `head()`, `info()`, and `describe()`.
 - Identified data types, summary statistics, and overall structure of the dataset.
 - Checked the total number of rows, columns, and unique values.
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3. Data Cleaning

- Checked for missing values to ensure data quality.
 - Removed the unnecessary 'Unnamed: 0' column from the dataset.
 - Verified that the dataset was clean and ready for analysis.
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4. Gender Distribution Analysis

- Analyzed the distribution of male and female students using a count plot.
 - Compared the proportion of each gender to understand the dataset composition.
 - Confirmed that the dataset contains a balanced representation of students.
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5. Parent's Education Analysis

- Calculated the average Mathematics, Reading, and Writing scores for each parental education level.
 - Created a heatmap to compare academic performance across education categories.
 - **Key Insight:** Students whose parents have Bachelor's or Master's degrees consistently achieved higher average scores, indicating that parental education positively influences academic performance.
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6. Parent's Marital Status Analysis

- Compared student performance across different parental marital status categories.
- Used a heatmap to visualize average subject scores.

- **Key Insight:** Student scores remained relatively consistent across all categories, suggesting that parental marital status has minimal impact on academic performance.
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7. Sports Participation Analysis

- Evaluated the relationship between sports participation and academic performance.
 - Compared average subject scores for students who regularly, sometimes, or never participate in sports.
 - **Key Insight:** Students who participate in sports regularly generally achieved better academic results, indicating a positive relationship between extracurricular activities and academic success.
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8. Subject-wise Score Distribution

- Created box plots for Mathematics, Reading, and Writing scores.
 - Analyzed score spread, median values, variability, and potential outliers.
 - **Key Insight:** Most students scored within a moderate range, while a few exceptional performers and low-performing outliers were identified.
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9. Ethnic Group Analysis

- Examined the distribution of students across different ethnic groups.
 - Visualized the demographic composition using count plots.
 - **Key Insight:** The dataset includes students from multiple ethnic backgrounds, providing a diverse sample for analysis.
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10. Key Business Insights

- Parental education is one of the strongest factors associated with higher academic performance.
 - Regular sports participation is positively associated with improved student scores.
 - Parental marital status has little measurable influence on academic achievement.
 - Score distributions indicate that while most students perform at an average level, a small group consistently outperforms others.
 - Data visualization effectively highlighted hidden trends that are difficult to identify from raw data alone.
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11. Conclusion

- Successfully transformed raw educational data into meaningful insights using Python.
- Identified the factors that have the greatest influence on student performance.
- Demonstrated an end-to-end data analysis workflow, from data cleaning to visualization and business insight generation, making this project suitable for a Data Analyst portfolio.